

Results: Per-User Heterogeneity and Sequential Structure

Combined dataset (Norway 100 + Korea 30 + Pecan 10 = 140 users, 28,159 sessions). Baselines: GMMNet (Li et al., 2024) and EV-SDG (Lahariya et al., 2020).

Per-user heterogeneity and sequential structure (combined, 140 users)

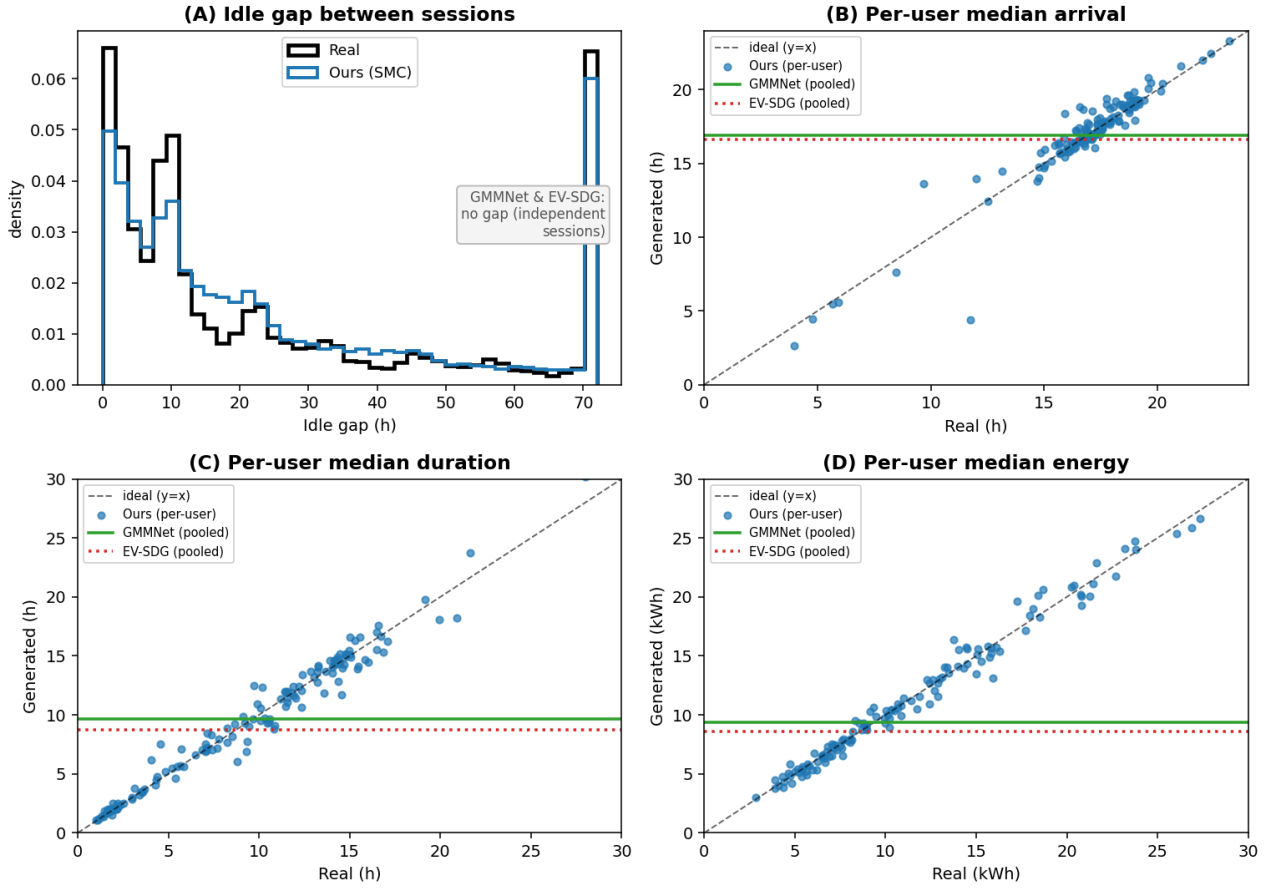


Fig. 1. (A) Only the proposed semi-Markov generator reproduces the idle gap between a user's consecutive sessions; the pooled baselines emit independent sessions and have no gap. (B-D) Per-user median of each variable, real (x) vs. generated (y): the proposed method (points) tracks the identity line, i.e. it reproduces each user individually, whereas the pooled baselines assign the same population value to every user (horizontal lines).

Quantitative comparison

Metric	Ours (SMC)	GMMNet	EV-SDG
Idle-gap KL (sequential)	0.036	nan	nan
Per-user arrival MAE (h)	0.5	1.69	1.78
Per-user duration MAE (h)	0.65	4.66	4.75
Per-user energy MAE (kWh)	0.66	6.17	6.29
Population marginal KL	0.008	0.012	1.533
Dependence (Kendall-tau err)	0.081	0.035	0.095

The green rows are the axes where the proposed method is uniquely or decisively better; the lower rows are reported for completeness. Idle-gap KL is N/A for pooled baselines (no sequential structure). Per-user MAE = mean over users of $|\text{median}(\text{real user}) - \text{median}(\text{generated})|$; the baselines have no per-user notion, so each user is scored against the single population value.

Reading of the results

- **Sequential structure (unique).** The proposed method reproduces the inter-session idle-gap distribution (KL 0.036); GMMNet and EV-SDG cannot represent it at all.
- **Per-user heterogeneity (decisive).** Per-user error is 3-9x lower than the pooled baselines: arrival 0.50 h vs 1.7 h, duration 0.65 h vs 4.7 h, energy 0.66 kWh vs 6.2 kWh. The baselines apply one population profile to every user (Fig.

1B-D, flat lines).

- **Population marginals (on par with SOTA).** All three are close; the neural GMMNet is competitive (0.012 vs 0.008), and EV-SDG is far worse only because its 24 h session cap discards residential multi-day sessions.
- **Dependence (baseline-competitive).** GMMNet edges the proposed method on Kendall-tau error; we do not claim superiority on static joint dependence.

Takeaway. On the marginal and dependence structure the proposed generator is on par with strong recent baselines, while it is the only method that captures *per-user heterogeneity* and the *sequential idle-gap* structure — the properties required for realistic, user-resolved, multi-session charging simulation.